

Kestrel

User Manual — version 2.3.0

Vantabrook Ltd. · Ashvale · August 2026

A fictional manual, published as sample material.

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1. Introduction

Kestrel is an invented task runner, and this manual is sample material used to exercise viewers, page counters, text extraction, and printing. Nothing in it documents a real product.

The manual is deliberately twenty pages long, with a table of contents, running headers, page numbers, and PDF outline entries, so that a tool reading it has something to find.

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2. Installation

Install the command globally, or run it once without installing. Both routes put the same binary on the path for the duration of the command.

There is nothing to configure at install time. Kestrel reads its configuration from the working directory when it runs.

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3. The configuration file

A project is described by a single `kestrel.toml` at its root. Each task declares the globs it reads, the paths it writes, the tasks it needs first, and the command to run.

Unknown keys are an error rather than a warning, because a silently ignored key is a cache key that quietly stops matching.

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4. Running tasks

Running a task runs everything it needs first, in dependency order, and in parallel where the graph allows it. A task whose inputs are unchanged is reported as cached and skipped.

Exit codes follow the usual convention: zero for success, one for a failed task, and two for a configuration error found before anything ran.

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5. The cache

The cache is content-addressed. A key is the hash of every declared input, the resolved command string, and the declared outputs. Timestamps are not part of it, so a fresh checkout still hits.

Entries are written to `.kestrel/cache` locally, and to the shared store when one is configured. A store miss and a key mismatch are counted separately, because the fixes for them have nothing in common.

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6. Watch mode

Watch mode re-runs a task when one of its declared inputs changes. Changes are debounced, and a run in progress is cancelled when a new change arrives.

Only declared inputs are watched. A task that reads a file it never declared will not re-run when that file changes, which is the same bug that makes it cache incorrectly.

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7. Troubleshooting

Start by printing the cache key for two consecutive runs and comparing them. A key that differs between identical checkouts points at an input that changes on its own, usually a generated file or a lockfile.

A key that differs only across machines points at the environment: a compiler version, a locale, or an absolute path that leaked into the output.

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8. Reference

Every command, flag, and configuration key, listed alphabetically. The tables here repeat what the earlier chapters describe in prose.

Where a default is given, it is the value used when the key is absent, not the value written by the initialiser.

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Appendix A. Configuration reference

Key	Type	Default	Description
inputs	string[]	[]	Globs whose contents decide the cache key
outputs	string[]	[]	Paths written by the task
needs	string[]	[]	Tasks that must succeed first
run	string	none	The shell command to execute
env	table	{}	Environment variables added to the command
cache	bool	true	Whether results may be cached at all
timeout	duration	30m	How long a task may run before it is killed

A key not listed here is an error, not a warning. Kestrel refuses to run rather than cache against a configuration it only partly understood.

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